

THE LANDFILL BOTS

IRL Build Roadmap — Two Arduino-Uno Robots

Parts · Design · Firmware · Assembly · Testing

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Page numbers are approximate — actual placement depends on final rendering.

1. Executive Summary

The Landfill Bots concept combines two jobs a real landfill site needs: cleaning up contaminated water (leachate/runoff) and sorting solid waste so more of it gets reused instead of buried. This roadmap turns that concept sketch into two independent, buildable robots that share a project name:

- **Bot 1 — Land Rover:** an 8-wheel-drive rover that detects waste with an IR sensor, picks it up with a 2-servo arm and a passive spring gripper, tests it (moisture + pH + color), and sorts it into wet/dry/plastic bins by rotating the arm to the right angle.
- **Bot 2 — Water Filter Unit:** a standalone, portable box that pumps contaminated water through a layered filter cartridge and outputs visibly clearer water — a demonstration of leachate treatment, not a drinking-water device.

Both run on Arduino Uno — one board per bot — which is a better fit for a first build than a single shared "smart" controller: each subsystem is simpler to wire, code, debug, and demo on its own, and if one board misbehaves on judging day, the other still works.

2. Design Notes — What Changed From the Sketch, and Why

A few details from the original concept got refined for physical buildability. Being able to explain these to judges is a strength, not a weakness — it shows you understand *why* the design works, not just what it looks like.

Why moisture + pH + color, not pH alone?

pH only means something on a wet sample — it can't tell you "this is plastic" by itself, since dry plastic and dry paper don't have a meaningful pH. The classification logic instead uses: **moisture sensor** to decide wet vs. dry (the physically correct test), **pH** as a secondary check when wet (decomposing organic waste trends acidic), and a **color/reflectance sensor** to separate plastic from dry organic waste when dry (plastics generally reflect more light than dull, matte organic matter). It's a heuristic, not a lab instrument — expect to tune thresholds against your actual waste samples.

Why 8-wheel-drive instead of tank tracks?

The sketch shows continuous tracks, but with 8 TT motors already on hand, wheels are the more natural fit — TT motors are designed for wheel hubs, not track sprockets. 8WD also gives excellent traction on uneven landfill-like terrain, is mechanically simpler to build and repair than tracks, and has no risk of a track derailing mid-demo.

Why does the arm's rotation double as the sorting mechanism?

With a passive spring gripper, both servos are free for pure positioning instead of one being spent on gripping. Mounting the arm on a rotating base means each bin can simply be a fixed angle — rotate to 20° for wet, 110° for dry, 160° for plastic — so no third "sorting gate" servo is needed at all.

The filtered water is a demonstration, not drinking water.

The filter cartridge reduces turbidity and captures debris/odour — a genuinely useful demo of the concept behind leachate treatment. It is **not** certified or intended for human consumption. Label the output container clearly during your demo.

3. System Architecture

The two bots share a design language but no wiring. Bot 1 handles everything land-based: driving, detecting, picking up, classifying, and sorting. Bot 2 is a simple, self-contained filtration appliance.

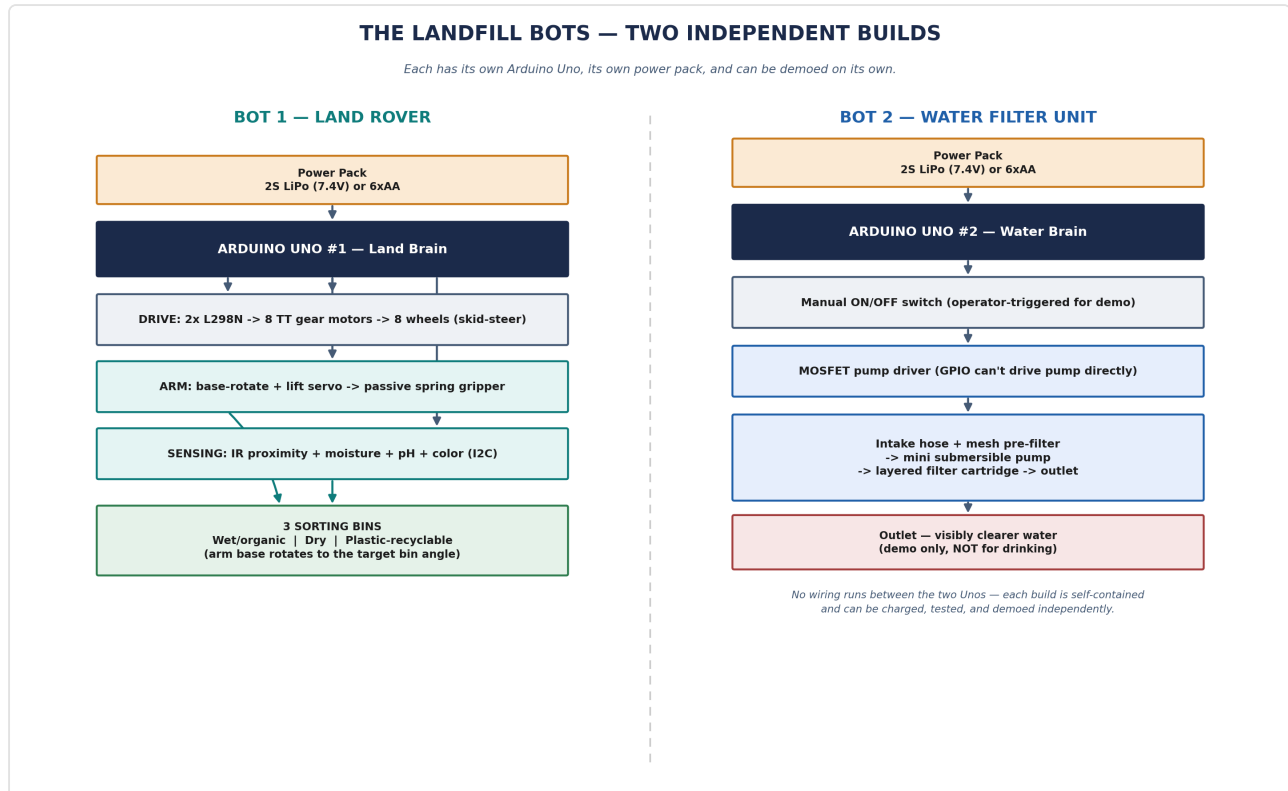


Figure 1 — System block diagram for both builds.

4. Bill of Materials

Budget isn't a constraint here, so this BOM favours reliable, well-supported parts over the cheapest option. Prices are approximate India street prices as of mid-2026 (Robu.in / Robocraze / ThingBits / Amazon.in / Flipkart) — expect $\pm 20\%$ depending on store and stock.

Already on hand — nothing to order

Item	Qty	Role in this build
Arduino Uno	2	Uno #1 = Land Rover brain, Uno #2 = Water Unit brain
L298N motor driver	2	Drive both sides of the 8-wheel chassis (4 motors/side, paired across boards)
TT gear motor	8	8-wheel-drive chassis, 4 per side
Servo motor	2	Arm base-rotation + lift (gripper is passive, no servo needed)

To order — Bot 1: Land Rover

Item	Qty	Approx. price	Notes
Wheels (TT-motor shaft fit)	8	₹40–60 ea (~₹400)	Skip if your TT motors already came with wheels
Chassis base plate	1	₹200–400	3–4mm acrylic or plywood, $\sim 22 \times 16$ cm
IR obstacle/proximity sensor module	1	₹40–80	Digital output, detects waste in gripper range
Capacitive soil moisture sensor	1	₹60–90	Capacitive, not resistive — won't corrode
Analog pH sensor kit (BNC probe + board)	1	₹1,500–2,200	DFRobot Gravity-style kit recommended for reliability
TCS34725 color sensor (I2C)	1	₹200–400	I2C only needs 2 pins — chosen over TCS3200 (needs 5) given Uno's limited pins
Spring-steel wire / compression springs (grripper prongs)	1 set	₹100–300	For your passive gripper mechanism
Acrylic/aluminium strip (arm boom + turret mount)	1 set	₹150–300	
Small bins/containers $\times 3$	3	₹150–300	Or repurpose containers — cost is flexible
Stripper-post hardware (bolts/rods at each bin)	3	₹50–100	Triggers passive gripper release — see \$5
2S LiPo battery (1300–2200mAh) + balance charger	1 + 1	₹900 + ₹600	Recommended. AA fallback: $6 \times$ AA holder, $\sim$ ₹150, no charger needed
Perfboard, jumper wires, screw terminals	1 set	₹250	
Mounting hardware (M3 screws, standoffs, servo horns, hot glue, cable ties)	1 set	₹300	
Toggle switch + power LED	1 each	₹80	

To order — Bot 2: Water Filter Unit

Item	Qty	Approx. price	Notes
Mini submersible DC water pump (3–6V)	1	₹150–250	
Silicone/PVC tubing	2m	₹100	Intake + outlet hose
Filter media: activated carbon, cotton wadding, fine mesh, sand/gravel	1 set	₹200	See \$9 for the layer order
Filter housing (PVC pipe section + end caps)	1	₹150–250	

			Or a repurposed clear plastic bottle for a visible demo
Logic-level MOSFET (e.g. IRLZ44N) or small relay module	1	₹40-100	Uno GPIO can't drive the pump directly
Toggle switch	1	₹40	Manual demo trigger
Small weatherproof project box	1	₹200-350	Houses Uno #2, battery, MOSFET
2S LiPo battery + charger (or share Bot 1's charger)	1	₹900 (+ shared charger)	Or 6×AA holder, ~₹150
Jumper wires, misc connectors	1 set	₹150	

Optional / nice-to-have

Item	Approx. price	Why
Multimeter	₹400-600	Essential for debugging wiring — get one if you don't have one
Hot glue gun	₹150-250	Fastest way to seal cable entries and secure the filter housing
0.96" I2C OLED display	₹200-300	Live status/classification readout during the demo

Approximate total to order: **₹5,800** (Land Rover) + **₹2,400** (Water Unit) $\approx$ **₹8,200**, plus optional extras. Verify current prices before ordering — component pricing shifts.

5. Mechanical Design & Structure

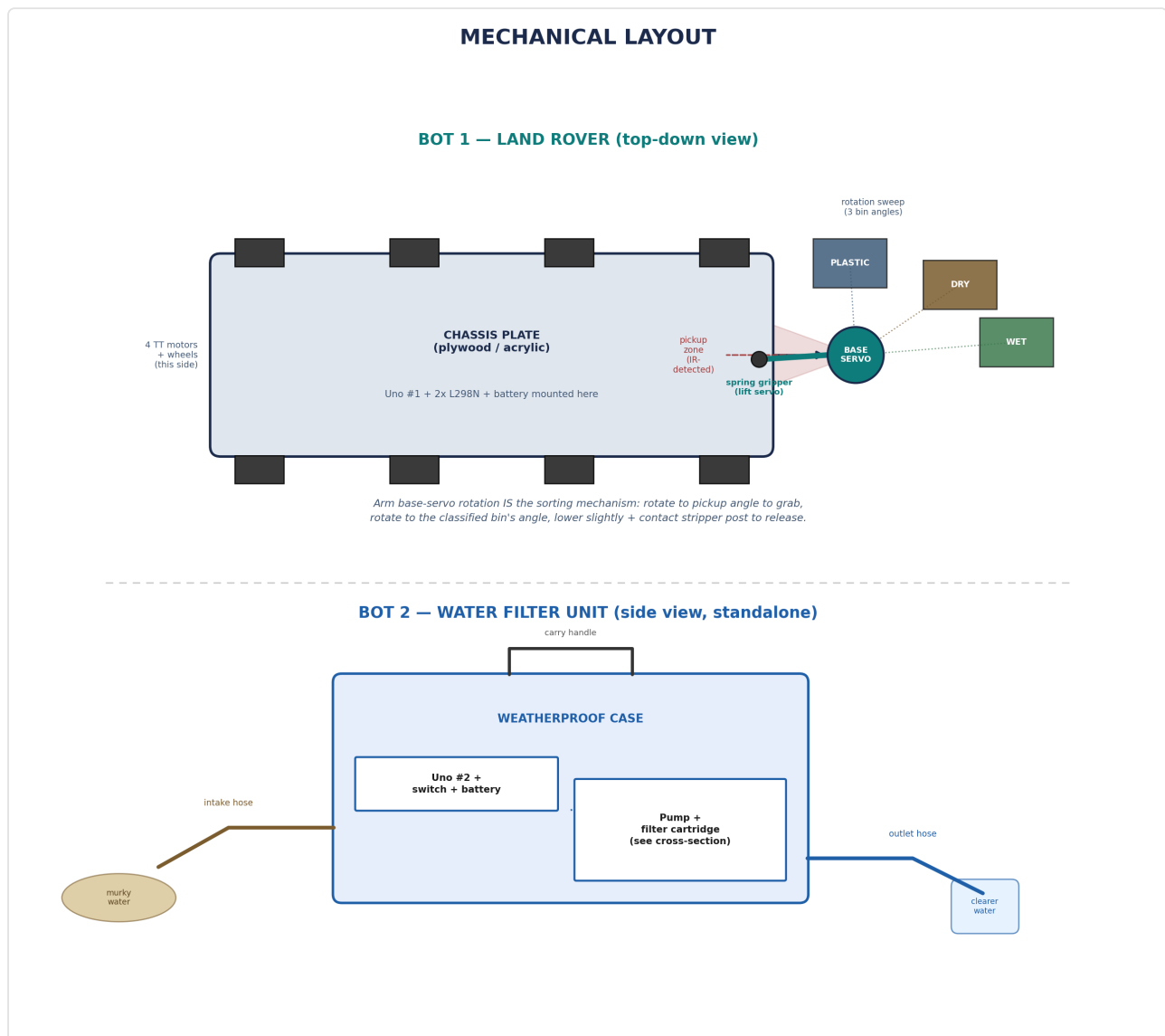


Figure 2 — Mechanical layout: Land Rover (top-down) and Water Filter Unit (side view).

Land Rover chassis

- Cut or laser-cut the base plate. Mount 4 TT motors per long edge, wheel shafts facing outward. Keep motors symmetric — mismatched mounting is the #1 cause of a rover that pulls to one side.
- Both L298N boards wire in parallel per side: left-front + left-rear share one channel, right-front + right-rear share the other. This isn't for independent per-wheel steering — it's simple skid-steer (left side / right side), doubled up across two boards for current headroom.
- Mount the arm's base servo on a small turret at one end of the plate, elevated enough that a 3-bin fan sits within its rotation sweep without hitting the chassis body.

Arm & passive gripper

- **Base servo** — horizontal rotation. Its angle doubles as the sort selector: one fixed angle per bin.
- **Lift servo** — raises/lowers the boom in a single vertical plane: down to grab, up to clear the bins while rotating.
- **Gripper** — your spring-loaded prongs, unpowered. As the lift servo lowers the boom onto a waste item, the prongs slide onto it and spring tension holds it.
- **Release** — the part most worth bench-testing early. The design here assumes a small fixed post ("stripper post") at each bin: as the arm lowers the held item toward the bin, the post pushes against the spring tension and cams the prongs open. If your specific prong geometry doesn't release cleanly against a post, a firm final push toward the bin

bottom is the simplest fallback — budget bench time for this specifically, since exact spring geometry is hard to predict on paper.

Water Filter Unit enclosure

- A small weatherproof project box holds Uno #2, battery, switch, and MOSFET — kept completely separate from the wet path.
- The pump and filter cartridge sit in their own compartment or are mounted external to the electronics bay, with cable entry points sealed (hot glue or a cable gland) so any drips can't reach the board.
- A handle on top makes it easy to carry between a demo water sample and the judging table.

6. Electrical Design & Wiring

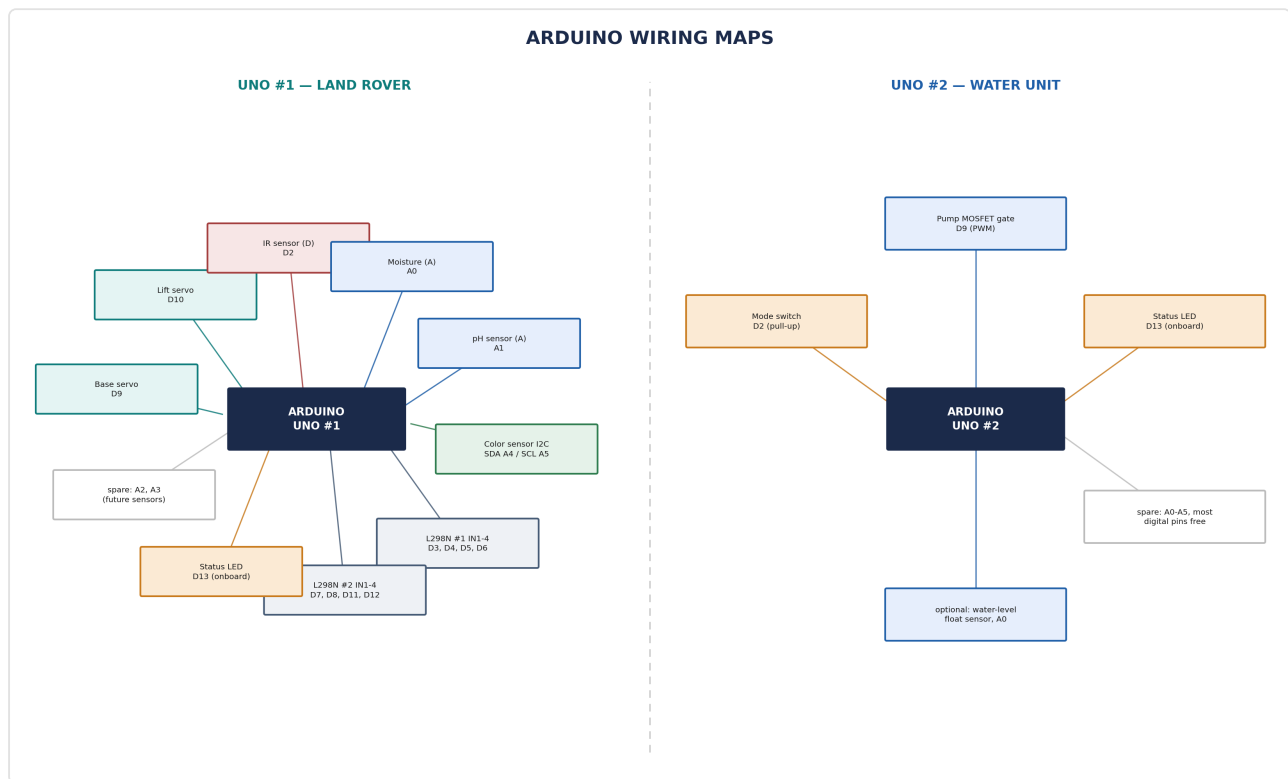


Figure 3 — GPIO assignment for both Unos.

Power architecture

Keep servo/motor power separate from logic power. Powering servos or motors directly off an Uno's 5V pin is the most common reason these builds behave erratically — voltage sags under load, and the Uno itself brownout-resets. Feed the Uno via its barrel jack/Vin (7–12V, regulated onboard to 5V), and feed servos + motors from the same battery pack but downstream of their own wiring, with a shared ground back to the Uno. If you see random resets when the arm moves, this is the first thing to check.

Pin reference — Uno #1 (Land Rover)

Signal	Pin	Signal	Pin
Base servo	D9	Moisture sensor (analog)	A0
Lift servo	D10	pH sensor (analog)	A1
IR sensor (digital)	D2	Color sensor SDA/SCL	A4 / A5
L298N #1 IN1–IN4	D3,D4,D5,D6	L298N #2 IN1–IN4	D7,D8,D11,D12
Status LED	D13 (onboard)	Spare	A2, A3

Pin reference — Uno #2 (Water Filter Unit)

Signal	Pin	Signal	Pin
Mode switch (pull-up)	D2	Pump MOSFET gate	D9
Status LED	D13 (onboard)	Spare	A0–A5, most digital pins

ENA/ENB on both L298N boards are jumpered HIGH on the board itself (fixed full-speed drive) — this frees up 4 pins that would otherwise be needed for PWM speed control. Add them back later if you want variable speed.

7. Firmware Architecture

Each Uno runs its own non-blocking state machine using `millis()` timing — no `delay()` calls, so the loop stays responsive. The two sketches share no code and no wiring; there is deliberately no serial link between them.

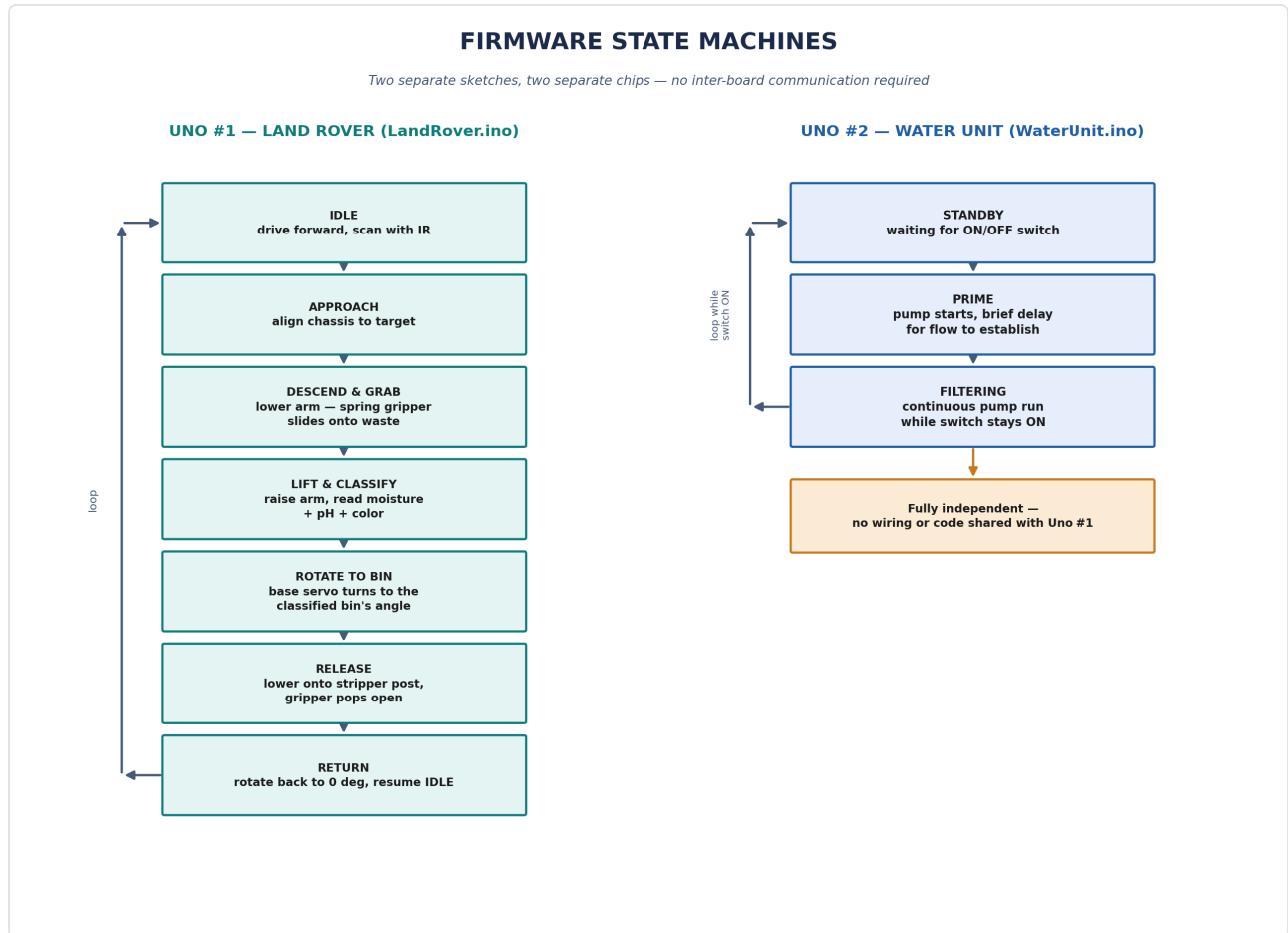


Figure 4 — State machines for both firmware sketches.

8. Firmware Source Code

Full source is also provided as ready-to-open .ino files in the accompanying firmware.zip — copy the folders straight into your Arduino sketchbook. LandRover.ino needs the **Adafruit TCS34725** library (Library Manager → search "Adafruit TCS34725").

Bot 1 — Land Rover

LandRover/LandRover.ino

```
#include <Servo.h>
#include <Wire.h>
#include "config.h"
#include "classify.h"

Servo baseServo;
Servo liftServo;

enum State { IDLE, APPROACH, DESCEND_GRAB, LIFT_CLASSIFY, ROTATE_TO_BIN, RELEASE, RETURN_HOME };
State state = IDLE;
unsigned long stateTimer = 0;
uint8_t targetBinAngle = BIN_WET_ANGLE;

void enterState(State s) { state = s; stateTimer = millis(); }

void setup() {
  Serial.begin(9600);
  Wire.begin();
  baseServo.attach(PIN_BASE_SERVO);
  liftServo.attach(PIN_LIFT_SERVO);
  baseServo.write(BASE_HOME_ANGLE);
  liftServo.write(LIFT_UP_ANGLE);
  pinMode(PIN_IR_SENSOR, INPUT);
  pinMode(PIN_L298N1_IN1, OUTPUT); pinMode(PIN_L298N1_IN2, OUTPUT);
  pinMode(PIN_L298N1_IN3, OUTPUT); pinMode(PIN_L298N1_IN4, OUTPUT);
  pinMode(PIN_L298N2_IN1, OUTPUT); pinMode(PIN_L298N2_IN2, OUTPUT);
  pinMode(PIN_L298N2_IN3, OUTPUT); pinMode(PIN_L298N2_IN4, OUTPUT);
  pinMode(LED_BUILTIN, OUTPUT);
  classifyInit();
  driveStop();
  Serial.println(F("LandRover ready."));
}

void loop() {
  switch (state) {
    case IDLE:
      driveForward(DRIVE_SPEED_SCAN);
      if (digitalRead(PIN_IR_SENSOR) == IR_TRIGGERED) { driveStop(); enterState(APPROACH); }
      break;
    case APPROACH:
      driveForward(DRIVE_SPEED_APPROACH);
      if (millis() - stateTimer > APPROACH_MS) { driveStop(); enterState(DSCEND_GRAB); }
      break;
    case DESCEND_GRAB:
      baseServo.write(BASE_PICKUP_ANGLE);
      liftServo.write(LIFT_DOWN_ANGLE); // spring gripper slides onto waste here
      if (millis() - stateTimer > DESCEND_MS) enterState(LIFT_CLASSIFY);
      break;
    case LIFT_CLASSIFY:
      liftServo.write(LIFT_UP_ANGLE);
      if (millis() - stateTimer > LIFT_MS) {
        WasteClass c = classifyItem();
        targetBinAngle = binAngleFor(c);
        Serial.print(F("Classified as: ")); Serial.println(wasteClassName(c));
        enterState(ROTATE_TO_BIN);
      }
      break;
    case ROTATE_TO_BIN:
      baseServo.write(targetBinAngle);
      if (millis() - stateTimer > ROTATE_MS) enterState(RELEASE);
      break;
    case RELEASE:
      liftServo.write(LIFT_RELEASE_ANGLE); // lowers onto stripper post
      if (millis() - stateTimer > RELEASE_MS) { liftServo.write(LIFT_UP_ANGLE); enterState(RETURN_HOME); }
      break;
    case RETURN_HOME:
      baseServo.write(BASE_HOME_ANGLE);
      if (millis() - stateTimer > RETURN_MS) enterState(IDLE);
      break;
  }
}
```

```

    digitalWrite(LED_BUILTIN, state == IDLE ? LOW : HIGH);
}

// Both L298N boards wired in parallel per side; ENA/ENB jumpered HIGH on
// the boards themselves for fixed full-speed operation.
void driveForward(int unusedSpeed) {
    digitalWrite(PIN_L298N1_IN1, HIGH); digitalWrite(PIN_L298N1_IN2, LOW);
    digitalWrite(PIN_L298N1_IN3, HIGH); digitalWrite(PIN_L298N1_IN4, LOW);
    digitalWrite(PIN_L298N2_IN1, HIGH); digitalWrite(PIN_L298N2_IN2, LOW);
    digitalWrite(PIN_L298N2_IN3, HIGH); digitalWrite(PIN_L298N2_IN4, LOW);
}

void driveStop() {
    digitalWrite(PIN_L298N1_IN1, LOW); digitalWrite(PIN_L298N1_IN2, LOW);
    digitalWrite(PIN_L298N1_IN3, LOW); digitalWrite(PIN_L298N1_IN4, LOW);
    digitalWrite(PIN_L298N2_IN1, LOW); digitalWrite(PIN_L298N2_IN2, LOW);
    digitalWrite(PIN_L298N2_IN3, LOW); digitalWrite(PIN_L298N2_IN4, LOW);
}

```

LandRover/config.h

```

#ifndef CONFIG_H
#define CONFIG_H

#define PIN_BASE_SERVO 9
#define PIN_LIFT_SERVO 10
#define PIN_IR_SENSOR 2
#define IR_TRIGGERED LOW // flip to HIGH if your IR module is active-HIGH
#define PIN_MOISTURE A0
#define PIN_PH A1
// Color sensor uses I2C: SDA=A4, SCL=A5 automatically via Wire.h

#define PIN_L298N1_IN1 3
#define PIN_L298N1_IN2 4
#define PIN_L298N1_IN3 5
#define PIN_L298N1_IN4 6
#define PIN_L298N2_IN1 7
#define PIN_L298N2_IN2 8
#define PIN_L298N2_IN3 11
#define PIN_L298N2_IN4 12

// Arm geometry (degrees) - TUNE ON THE BENCH before first run
#define BASE_HOME_ANGLE 90
#define BASE_PICKUP_ANGLE 90
#define BIN_WET_ANGLE 20
#define BIN_DRY_ANGLE 110
#define BIN_PLASTIC_ANGLE 160
#define LIFT_UP_ANGLE 30
#define LIFT_DOWN_ANGLE 150
#define LIFT_RELEASE_ANGLE 140

// Timing (ms) - non-blocking, tune to taste
#define APPROACH_MS 600
#define DESCEND_MS 700
#define LIFT_MS 500
#define ROTATE_MS 600
#define RELEASE_MS 500
#define RETURN_MS 500
#define DRIVE_SPEED_SCAN 255 // reserved for a future PWM upgrade
#define DRIVE_SPEED_APPROACH 255

#endif

```

LandRover/classify.h

```

#ifndef CLASSIFY_H
#define CLASSIFY_H
#include <Adafruit_TCS34725.h>
#include "config.h"

enum WasteClass { WASTE_WET, WASTE_DRY, WASTE_PLASTIC };
Adafruit_TCS34725 tcs = Adafruit_TCS34725(TCS34725_INTEGRATIONTIME_50MS, TCS34725_GAIN_4X);

#define MOISTURE_WET_THRESHOLD 550 // 0-1023 - calibrate: dry sample vs wet sample
#define REFLECTANCE_PLASTIC_MIN 180 // TCS34725 clear channel - calibrate on the bench

void classifyInit() {
    if (!tcs.begin()) Serial.println(F("WARNING: TCS34725 not found - check wiring."));
}

int readMoisture() {
    long sum = 0;
    for (int i = 0; i < 8; i++) { sum += analogRead(PIN_MOISTURE); delay(2); }
}

```

```

    return sum / 8;
}
float readPH() {
    int raw = analogRead(PIN_PH);
    float voltage = raw * (5.0 / 1023.0);
    return 7.0 - ((voltage - 2.5) / 0.18); // replace with your 2-point calibration
}
uint16_t readReflectance() {
    uint16_t r, g, b, c;
    tcs.getRawData(&r, &g, &b, &c);
    return c;
}
WasteClass classifyItem() {
    int moisture = readMoisture();
    if (moisture > MOISTURE_WET_THRESHOLD) {
        float ph = readPH();
        Serial.print(F("moisture=")); Serial.print(moisture);
        Serial.print(F("  pH=")); Serial.println(ph);
        return WASTE_WET;
    }
    uint16_t reflectance = readReflectance();
    Serial.print(F("moisture=")); Serial.print(moisture);
    Serial.print(F("  reflectance=")); Serial.println(reflectance);
    return (reflectance > REFLECTANCE_PLASTIC_MIN) ? WASTE_PLASTIC : WASTE_DRY;
}
uint8_t binAngleFor(WasteClass c) {
    switch (c) {
        case WASTE_WET: return BIN_WET_ANGLE;
        case WASTE_PLASTIC: return BIN_PLASTIC_ANGLE;
        default: return BIN_DRY_ANGLE;
    }
}
const char* wasteClassName(WasteClass c) {
    switch (c) {
        case WASTE_WET: return "WET / ORGANIC";
        case WASTE_PLASTIC: return "PLASTIC";
        default: return "DRY";
    }
}
}
#endif

```

Bot 2 — Water Filter Unit

WaterUnit/WaterUnit.ino

```

#define PIN_MODE_SWITCH 2 // to GND, uses internal pull-up
#define PIN_PUMP_GATE 9 // to MOSFET gate driving the pump

enum State { STANDBY, PRIME, FILTERING };
State state = STANDBY;
unsigned long stateTimer = 0;
#define PRIME_MS 1500

void setup() {
    Serial.begin(9600);
    pinMode(PIN_MODE_SWITCH, INPUT_PULLUP);
    pinMode(PIN_PUMP_GATE, OUTPUT);
    pinMode(LED_BUILTIN, OUTPUT);
    digitalWrite(PIN_PUMP_GATE, LOW);
    Serial.println(F("WaterUnit ready."));
}

void loop() {
    bool switchOn = (digitalRead(PIN_MODE_SWITCH) == LOW);
    switch (state) {
        case STANDBY:
            digitalWrite(PIN_PUMP_GATE, LOW);
            digitalWrite(LED_BUILTIN, LOW);
            if (switchOn) {
                digitalWrite(PIN_PUMP_GATE, HIGH);
                stateTimer = millis();
                state = PRIME;
            }
            break;
        case PRIME:
            digitalWrite(LED_BUILTIN, (millis() / 150) % 2);
            if (!switchOn) { state = STANDBY; break; }
            if (millis() - stateTimer > PRIME_MS) state = FILTERING;
            break;
        case FILTERING:
            digitalWrite(LED_BUILTIN, HIGH);
            if (!switchOn) state = STANDBY;

```

```
    break;  
  }  
}
```

9. Water Filter Cartridge Detail

WATER FILTER CARTRIDGE — CROSS-SECTION

Demonstrates turbidity / contaminant reduction — outlet water is NOT for drinking

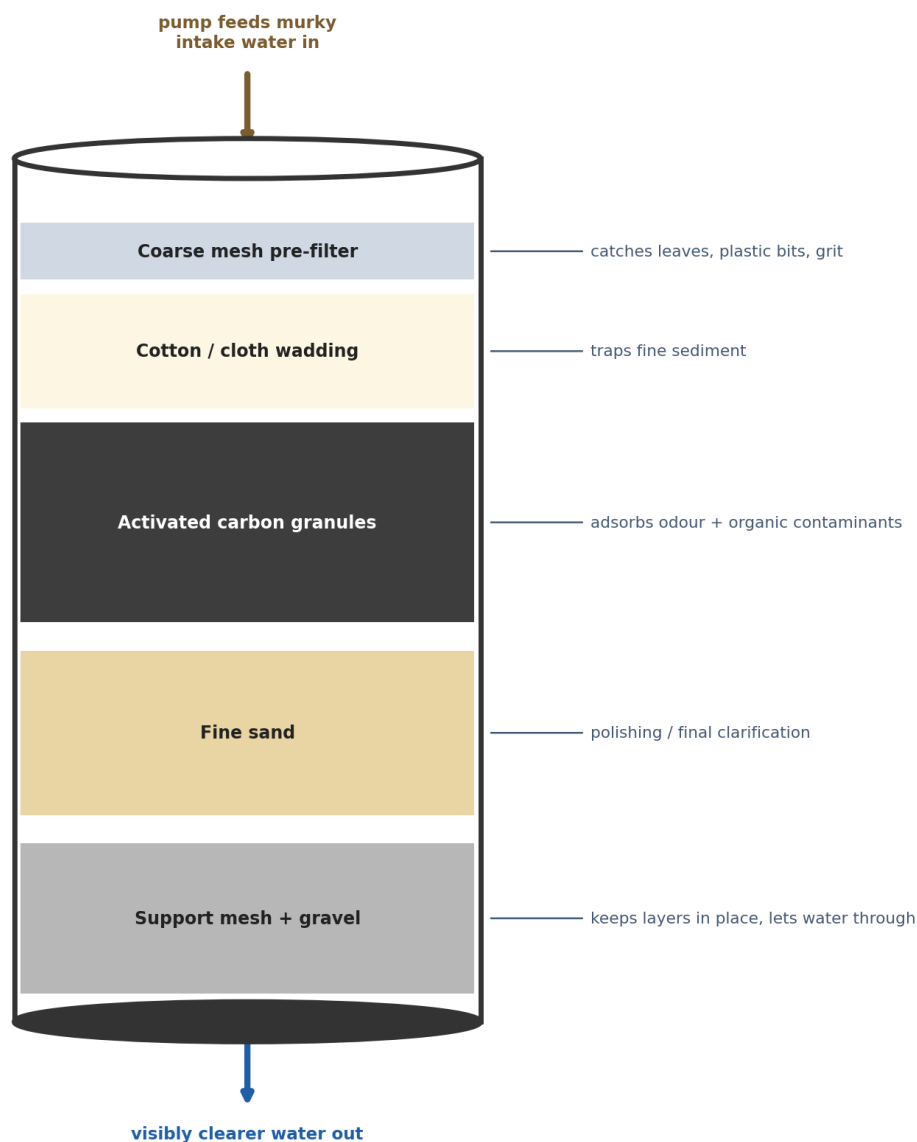


Figure 5 — Filter cartridge layer order.

Build the cartridge in a length of PVC pipe (or a clear plastic bottle with the bottom cut off, for a more visible demo). Layer order matters — coarse to fine, top to bottom, so the pre-filter catches large debris before it can clog the finer layers below:

1. **Coarse mesh** — catches leaves, plastic fragments, grit
2. **Cotton/cloth wadding** — traps fine suspended sediment
3. **Activated carbon granules** — adsorbs odour and organic contaminants
4. **Fine sand** — final polishing/clarification
5. **Support mesh + gravel** — holds everything in place, lets water through to the outlet

10. Assembly Procedure

Bot 1 — Land Rover

1. Cut/prepare the chassis base plate. Mark and mount all 8 TT motors symmetrically (4 per side).
2. Press-fit or screw on all 8 wheels. Spin each motor by hand-connecting to battery briefly to confirm free rotation before wiring anything permanent.
3. Mount both L298N boards on the plate; wire IN1-IN4 per board to the Uno per the pin table in §6. Wire left-side motors in parallel to one L298N channel, right-side to the other, mirrored across both boards.
4. Bench-test drive: upload a short forward/stop/backward test sketch before installing the arm. Confirm both sides drive the same direction for "forward" — swap motor leads on any wheel spinning backward.
5. Build the arm: mount the base servo on its turret, attach the boom to the lift servo, fit your spring gripper at the boom tip.
6. Mount the IR sensor facing the pickup zone, angled so it triggers when an item is within gripper reach.
7. Wire the moisture sensor, pH probe, and TCS34725 per §6. Mount the pH probe and moisture sensor where the gripper naturally rests when lifted (post-pickup), so both can contact/sense the held item.
8. Build and position the 3 bins at the base-servo angles set in `config.h`. Fit each with a stripper post at the height the lowered gripper would contact.
9. Wire power: battery → Uno (Vin/barrel jack), battery → servos and L298N boards (shared ground back to the Uno).
10. Upload `LandRover.ino`. Run through the state machine manually via Serial Monitor prints before the full autonomous loop.

Bot 2 — Water Filter Unit

1. Assemble the filter cartridge per §9's layer order inside your chosen housing.
2. Mount the pump at the housing's base/inlet side; connect intake tubing with the coarse mesh pre-filter at its open end.
3. Connect outlet tubing from the cartridge's far end to a spout or collection point.
4. Wire the pump through the MOSFET (or relay) rather than directly to a Uno pin — GPIO pins can't supply a pump's current.
5. Wire the toggle switch to D2 and the MOSFET gate to D9 per §6.
6. Mount Uno #2, battery, and MOSFET in the weatherproof case, physically separated from the wet path. Seal cable entry points.
7. Upload `WaterUnit.ino`. Test standalone with a bucket of dirty water before combining with the rest of the presentation.

11. Testing & Calibration

Moisture threshold

Read `analogRead(PIN_MOISTURE)` on a bone-dry item and a soaking-wet item via Serial Monitor. Set `MOISTURE_WET_THRESHOLD` roughly halfway between the two readings.

pH sensor calibration

Use pH 4.0 and pH 7.0 buffer solutions (available cheaply alongside the sensor kit). Record the voltage reading at each, then compute the actual slope/offset for the line in `readPH()` — the values in the code are a generic starting point, not a substitute for calibrating your specific probe.

Color sensor / plastic threshold

Log the `cClear` channel reading for a few dry plastic items and a few dry organic items (dry leaves, paper). Set `REFLECTANCE_PLASTIC_MIN` between the two clusters. Test under consistent lighting — ambient light changes will shift readings.

Arm angles & gripper release

Tune `BASE_PICKUP_ANGLE`, the three bin angles, and both lift angles by direct trial with the arm powered but the rover stationary. This is also where you'll dial in the stripper-post height and position for a reliable passive release — budget real bench time here, since it depends on your exact gripper geometry.

Integration test order

1. Drive system alone (no arm powered)
2. Arm alone (rover stationary, manually trigger each state via Serial commands if useful)
3. Sensors alone (log readings for known wet/dry/plastic samples)
4. Full autonomous loop, supervised, with an easy power cutoff within reach
5. Water Unit, independently, with a bucket of intentionally murky water

12. Build Timeline

Week 1 — Order parts. Build & bench-test the drive chassis (8WD, both L298N boards).

Week 2 — Build the arm + passive gripper. Tune base/lift angles and the release mechanism.

Week 3 — Wire and calibrate sensors (IR, moisture, pH, color). Build the 3-bin sorting station.

Week 4 — Integrate Land Rover firmware end-to-end; debug the full pickup→classify→sort loop.

Week 5 — Build the Water Filter Unit in parallel (independent, so this can overlap earlier weeks if you have a second pair of hands).

Week 6 — Polish, weatherproof, rehearse the demo, prepare the presentation board.

13. Safety Notes

LiPo batteries: only charge with a proper balance charger, never leave charging unattended, store in a fireproof LiPo bag, and never puncture, short, or charge a swollen cell.

Water + electronics: keep the Water Unit's electronics bay physically sealed from the wet path. Test for leaks with plain water before wiring anything live.

Moving parts: the spring gripper and rotating arm are pinch-point hazards — keep fingers clear during testing, and consider a simple guard/shroud around the gripper for the live demo.

Filtered water is not for drinking. Label the output container during any demo.